

D301

Developing an E-Learning Solution and Research Methodology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A design team meets in a bright studio to ideate solutions for adult learners who struggle with a new expense system. The first person suggests a chatbot; the lead immediately says it is too costly. The team has three weeks left in the term. What principle is being broken?

- A. Converge on one idea quickly because of the deadline.
- B. Always pick the cheapest idea first.
- C. Test the chatbot with users before discussing other ideas.
- D. Defer judgment and generate many ideas before narrowing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A researcher uses a 20-item survey to measure adult learners' ability to use spreadsheet formulas. Scores are very consistent when the same learners retake it a week later. The survey was printed on recycled paper and takes ten minutes. However, it only asks learners to rate their own confidence. What is the main measurement concern?

- A. It is too short at ten minutes.
- B. Reliability: scores may change from week to week.
- C. It is valid because it is reliable.
- D. Validity: it measures confidence, not formula skill.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A designer's research question is: "How do adult learners describe their experience navigating the new onboarding module?" The module was built in six weeks and includes 14 videos. Which approach fits best?

- A. Count how many of the 14 videos each learner watched.
- B. Qualitative interviews or focus groups analyzed for themes.
- C. Quantitative methods only, because numbers are always more rigorous.
- D. A pre/post test comparing scores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A researcher plans to analyze interviews coded for themes. Which research question would this method answer best?

- A.** How much did quiz scores rise after the module?
- B.** Why do some adult learners stop partway through the module?
- C.** Is there a statistically significant difference between two groups?
- D.** What percentage of learners completed the module?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Adult learners in a single cohort took a pre-test on workplace writing, used a new e-learning module for eight weeks, and took a post-test. Scores rose 12 points. During those weeks, the company also required all staff to use a new writing template. The module has a clean, modern interface. What is the strongest reason to be cautious about crediting the module?

- A.** The modern interface may have distracted learners.
- B.** There is no concern; a pre/post gain proves the module worked.
- C.** A 12-point gain is too small to matter.
- D.** The new template is a history threat; without a comparison group, the gain cannot be credited to the module alone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An instructor who teaches an online certificate course wants to study her own learners' use of a new e-learning module. The course has 40 adults from several industries and meets on Thursday evenings. Which recruitment plan best protects voluntary consent?

- A.** Make participation a course requirement so the sample is complete.
- B.** Recruit only on Thursday evenings, when everyone is present.
- C.** Have a third party handle consent and keep the list of participants hidden from the instructor until grades are final.
- D.** Give 5 extra-credit points only to those who agree to participate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A participant emails halfway through a study to say she wants to stop. What should the researcher do?

- A.** Accept the withdrawal without penalty and follow the consent form on handling her data.
- B.** Accept, but note it in her course record.
- C.** Keep collecting her data quietly for completeness.
- D.** Ask her to finish, since she already signed consent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A designer plans an anonymous survey of adult learners about an e-learning module. He works at a large university with a busy research office and has conducted studies before. He believes the survey is harmless. What should he do before collecting data?

- A.** Skip review because anonymous surveys never need it.
- B.** Collect the data first and submit for review if a problem arises.
- C.** Submit the plan to the ethics review board and let it determine whether the study is exempt.
- D.** Declare the study exempt himself, since he has done studies before.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A researcher collects survey responses with each learner's email address so she can match pre- and post-surveys. She stores the file on an encrypted drive in a locked office. Her report describes the data as anonymous. The survey took about 12 minutes. What is wrong?

- A.** The survey is too long to be anonymous.
- B.** Nothing; encryption makes the data anonymous.
- C.** The data are confidential, not anonymous, because responses are linked to email addresses.
- D.** The data are anonymous because they are in a locked office.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A researcher wants responses that are truly anonymous. Which collection method achieves this?

- A.** An online survey that collects no names, emails, IDs, or IP addresses.
- B.** A survey with names that the researcher deletes after analysis.
- C.** Interviews recorded on video and stored securely.
- D.** A survey using learner ID numbers instead of names.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A designer wants to test an e-learning module with 16-year-old students in a summer program run by a nonprofit that has operated for 20 years. The program meets in a leased office building. Which consent process is appropriate?

- A.** Get a parent or guardian's permission and each student's own assent.
- B.** Get the students' assent only, since they are 16.
- C.** Get the nonprofit director's approval only.
- D.** Get guardian permission only; the students' views do not matter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A parent signs a permission form for a 14-year-old to join a study, but the teen says she does not want to take part. What should the researcher do?

- A.** Ask the parent to persuade her until she agrees.
- B.** Do not enroll her; her refusal to assent is respected.
- C.** Enroll her, because the parent's permission is legally sufficient.
- D.** Enroll her but keep her data confidential.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A design team ~~meets in a bright studio~~ to ideate solutions for adult learners who struggle with a new expense system. The first person suggests a chatbot; the lead immediately says it is too costly. ~~The team has three weeks left in the term.~~ What principle is being broken?

Target word: ideation rule broken **Decides:** An idea is judged as soon as it is proposed.

A	Camouflage	The deadline does not change the diverge-first rule.
B	Absolute language	Cost is a later criterion, not a rule for ideation.
C	Adjacent category	Testing is a later phase; ideation comes first.
D	Correct	Early criticism shuts down divergent thinking.

2. Answer D Concept 2

A researcher uses a 20-item survey to measure adult learners' ability to use spreadsheet formulas. Scores are very consistent when the same learners retake it a week later. ~~The survey was printed on recycled paper and takes ten minutes.~~ However, it only asks learners to rate their own confidence. What is the main measurement concern?

Target word: validity or reliability problem **Decides:** Self-rated confidence does not measure actual ability.

A	Camouflage	Length is not the issue here.
B	Confusion set	The stem says scores are consistent.
C	Inverted operation	Reliability is needed for validity but does not guarantee it.
D	Correct	It is consistent (reliable) but does not measure the stated construct.

3. Answer B Concept 3

A designer's research question is: "How do adult learners describe their experience navigating the new onboarding module?" ~~The module was built in six weeks and includes 14 videos.~~ Which approach fits best?

Target word: method for the question **Decides:** An experience question ("how do learners describe").

A	Camouflage	Uses a stem number that does not answer the question.
B	Correct	Descriptions of experience call for qualitative data.
C	Absolute language	Rigor depends on fit to the question.
D	Adjacent category	Measures change in scores, not experience.

4. Answer B Concept 3 · transfer

A researcher plans to analyze interviews coded for themes. Which research question would this method answer best?

Target word: reverse: pick the question for the method **Decides:** Thematic interview analysis fits how/why questions.

A	Inverted operation	A how-much question needs quantitative data.
B	Correct	A why question about experience suits interviews.
C	Adjacent category	Needs a statistical test, not thematic coding.
D	Confusion set	A completion rate comes from system data.

5. Answer D Concept 4

Adult learners in a single cohort took a pre-test on workplace writing, used a new e-learning module for eight weeks, and took a post-test. Scores rose 12 points. During those weeks, the company also required all staff to use a new writing template. ~~The module has a clean, modern interface.~~ What is the strongest reason to be cautious about crediting the module?

Target word: threat to the causal claim **Decides:** A concurrent event (the template) could explain the gain; no comparison group.

A	Camouflage	Interface style does not undermine the causal claim.
B	Absolute language	Pre/post gains alone cannot rule out other causes.
C	Wrong stem number	The size of the gain is not the flaw; the design is.
D	Correct	An outside event during the study could cause the gain.

6. Answer C Concept 5

An instructor who teaches an online certificate course wants to study her own learners' use of a new e-learning module. The course has 40 adults ~~from several industries and meets on Thursday evenings.~~ Which recruitment plan best protects voluntary consent?

Target word: voluntary consent with a power relationship **Decides:** The instructor grades the participants.

A	Absolute language	Required participation is not voluntary.
B	Camouflage	Timing does not address the power problem.
C	Correct	Removes pressure; learners can decline without fear of grade effects.
D	Inverted operation	Tying grades to participation creates pressure.

7. Answer A Concept 5 · transfer

A participant emails halfway through a study to say she wants to stop. What should the researcher do?

Target word: right to withdraw **Decides:** Withdrawal is allowed at any time without penalty.

A	Correct	Participants can withdraw at any time without penalty.
B	Adjacent category	Any record that could affect her is a penalty.
C	Dropped element	Ignores her withdrawal entirely.
D	Inverted operation	Consent is ongoing and can be withdrawn.

8. Answer C Concept 6

A designer plans an anonymous survey of adult learners about an e-learning module. He works at a large university ~~with a busy research office and has conducted studies before.~~ He believes the survey is harmless. What should he do before collecting data?

Target word: who decides exemption **Decides:** The review board makes that call.

A	Absolute language	Anonymous studies may be exempt, but the board decides.
B	Inverted operation	Review must come before data collection.
C	Correct	Exempt status is decided by the board, not the researcher.
D	Camouflage	Experience does not give authority to self-exempt.

9. Answer C Concept 7

A researcher collects survey responses with each learner's email address so she can match pre- and post-surveys. She stores the file on an encrypted drive in a locked office. Her report describes the data as anonymous. ~~The survey took about 12 minutes.~~ What is wrong?

Target word: anonymous vs. confidential **Decides:** Emails link responses to people.

A	Camouflage	Length has nothing to do with identifiability.
B	Confusion set	Encryption protects confidentiality; it does not remove identity.
C	Correct	If the researcher can identify a person, the data are not anonymous.
D	Adjacent category	Physical security is a confidentiality safeguard.

10. Answer A Concept 7 · transfer

A researcher wants responses that are truly anonymous. Which collection method achieves this?

Target word: reverse: design for anonymity **Decides:** No identifiers collected at all.

A	Correct	With no identifiers, no one can link data to a person.
B	Inverted operation	Data were identifiable during the study; that is confidentiality.
C	Adjacent category	Video identifies people; it can only be confidential.
D	Confusion set	IDs can be linked back to people.

11. Answer A Concept 8

A designer wants to test an e-learning module with 16-year-old students in a summer program run by a nonprofit ~~that has operated for 20 years. The program meets in a leased office building.~~ Which consent process is appropriate?

Target word: consent for minors **Decides:** Participants are under 18.

A	Correct	Minors need both guardian permission and their own agreement.
B	Dropped element	Leaves out guardian permission.
C	Adjacent category	Site permission does not replace guardian permission and student assent.
D	Absolute language	Minors' assent is still required.

12. Answer B Concept 8 · transfer

A parent signs a permission form for a 14-year-old to join a study, but the teen says she does not want to take part. What should the researcher do?

Target word: reverse: assent refused **Decides:** The minor's assent is also required.

A	Inverted operation	Pressure undermines voluntary participation.
B	Correct	Guardian permission alone is not enough when the minor declines.
C	Dropped element	Drops the assent requirement.
D	Adjacent category	Confidentiality does not replace assent.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Diverge before converge	Ideation generates many options with judgment deferred, then narrows with criteria.	Evaluating or dismissing ideas as they are proposed.	A brainstorm where the first idea is critiqued immediately.
2	Validity vs. reliability	Validity asks whether an instrument measures what it claims; reliability asks whether it measures consistently.	Assuming a consistent measure must be a valid one.	Stable scores on an instrument that measures the wrong thing.
3	Match the method to the question	“How much” or “whether” questions call for quantitative data; “how” and “why” experience questions call for qualitative data; both call for mixed methods.	Choosing a method before writing the question.	The question words: how much, did it change, why, how do learners experience.
4	Threats in a one-group pre/post design	Without a comparison group, gains may come from maturation, history, or testing effects rather than the intervention.	Reading any pre/post gain as proof the solution worked.	A single group tested before and after, with other events in between.
5	Voluntary informed consent	Participants must understand the study and be free to decline or withdraw without penalty, especially when the researcher has power over them.	Treating course requirements or grades as a way to get participation.	An instructor or supervisor recruiting their own learners.
6	Who decides review status	An ethics review board, not the researcher, determines whether a human-subjects study is exempt or needs full review.	Self-declaring a study exempt because it seems harmless.	A researcher deciding on their own that review is unnecessary.
7	Confidential vs. anonymous data	Anonymous means no one, including the researcher, can link data to a person; confidential means the researcher can link it but protects it.	Calling data anonymous when names or IDs were collected.	Data collected with names or IDs described as anonymous.
8	Extra protection for minors	Research with minors needs parent or guardian permission plus the child’s own assent.	Thinking a parent’s permission alone is enough, or that a minor’s agreement alone is enough.	Participants under 18.

Course scope (public description): Developing an E-Learning Solution and Research Methodology is the second of three capstone courses in the program. This course focuses on the next two phases of the design-based research process: designing and developing an e-learning solution and designing a research methodology to test how well the solution addressed the instructional problem. This course also requires that learners continue applying Design Thinking as they ideate potential solutions to the instructional problem and begin prototyping a module of instruction. Finally, this course teaches learners how to design research studies that ensure the safety of human subjects and the ethical collection, storage, and reporting of data. The course Identifying Learner Needs and a Research Problem is a prerequisite for this course.